ABDULREHMAN OMER

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Software Development Engineer with 3+ years of non-internship professional software development experience and 2+ years of non-internship design or architecture (design patterns, reliability and scaling) experience, programming in the Python software programming language, with full software development life cycle experience (coding standards, code reviews, source control, build processes, testing, operations) and hands-on work building distributed data pipelines and productionizing ML models at scale.

Education

Memorial University of Newfoundland **Sep 2019 - Apr 2023**
Bachelor of Science | Major in Computer Science, Minor in Biochemistry *St. John's, NL*

Work Experience

Bizperiscope Inc. **Mar 2026 - Present**
Software Engineer - AI/ML *London, ON*

- Design, build, and operate **distributed data pipelines** processing large-scale event and panel-style data over **Databricks**, **Delta Lake**, and **Apache Spark (PySpark)**, deploying on **Azure** and **AKS** (**Kubernetes**).
- Partner on **ML model productionization**: build the backend services and data contracts in **Python** that take a model from prototype to a production-grade system serving live decisions.
- Create and monitor **production metrics** for system **availability, reliability, and performance**, improving observability to catch silent failures, and carry **on-call** responsibility for incident response on live services.

Prime Communications Canada **Jun 2025 - Feb 2026**
Senior Technical Lead *London, ON*

- Partnered with account executives and engineering teams to design customized technical solutions, delivering technical presentations and **proof-of-concept** sessions that translated ambiguous business needs into scalable, cost-effective systems.

Rogers Communications **Apr 2023 - May 2025**
Software Engineer (L1 to L2) *Toronto, ON*

- Owned end-to-end **design, development, and launch** of backend systems and data pipelines processing high transaction volumes in **Python** (**Django/FastAPI**), applying **design patterns** for reliability and scaling, progressing from contributor (L1) to feature owner (L2).
- Led technical design reviews and **mentored junior and incoming engineers** through code reviews, raising the engineering bar and establishing best practices across the team.
- Navigated a large shared codebase to own **system integration testing** and debug/resolve defects across multiple features under production load, writing clear, well-tested, extensible code and reducing post-QA escalations and repeat incidents.
- Worked across relational and non-relational databases (**Oracle**, **PostgreSQL**, **MongoDB**) and strengthened **CI/CD** pipelines using **Git**, **Jenkins**, and **Docker** through the full software development life cycle.

Canadian Tire Corporation **Jan 2022 - Apr 2022**
Software Engineer Intern *Toronto, ON (Hybrid)*

- Designed and maintained **ETL pipelines** on **Azure** processing large-scale datasets, and collaborated with analysts to ensure data availability and accuracy.

Projects

Clinical Intelligence Platform - Medical RAG Chatbot (Agentic) **Jul 2026 - Present**
Python | LangChain | LangGraph | FAISS | Hugging Face | FastAPI

- Built an **agentic, LLM-powered** application: retrieved relevant context from a **FAISS** vector store and generated grounded, evidence-based answers, served through a **FastAPI** backend, tuning for **accuracy, latency, and cost**.

Certifications

- AWS: **AWS Certified Cloud Solutions Architect** | IBM: **RAG and Agentic AI** · **Generative AI Engineering** · **Full Stack Software Developer** · **Machine Learning Specialization** (2026) | Packt: **Solution Architecture and Cloud Design Specialization** · **System Design & Architecture**

Technical Skills

Languages: Python, Java (working knowledge), JavaScript, TypeScript, SQL, HTML5, CSS
Big Data & Pipelines: Apache Spark (PySpark), Databricks, Delta Lake, distributed data pipelines, large-scale event processing
System & API Design: Design patterns, distributed systems, scalability, reliability, maintainability, extensibility, and performance trade-offs, REST APIs
ML & AI: ML model productionization, LLM-based agents, RAG, ML model integration (XGBoost), prompt engineering
Operational Excellence: Production monitoring, observability, on-call incident response, root-cause analysis, full SDLC (code review, source control, build/test/deploy)
Cloud & DevOps: AWS (S3, Lambda) - AWS Certified Cloud Solutions Architect, Azure, Kubernetes (AKS), Docker, Git, CI/CD, Azure DevOps, Jenkins
Databases: PostgreSQL, Oracle, MongoDB